



A good plan goes a long way

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Apprentice cryptographer, my main research areas are coding theory and protocol security. My interests include scientific communication and outreach. I always strive to create.

TECHNICAL SKILLS

Programming:

Python, Magma, C, Java, R.

Web:

React, Angular.

Typesetting:

LaTeX, Beamer.

SOFT SKILLS

Developed through research, industry, and scientific event organisation: structured planning, adaptability across technical contexts, and collaborative leadership.

SPOKEN LANGUAGES

Italian: Native

English: Professional proficiency

German: Basic

AWARDS

Cyprus Tech Odyssey 2024
Hackathon, 1st Place

Matematica in gioco 2023
Archimede Prize, Venice

Catch The Mark 2022
Watermarking challenge, 1st Place

OUTREACH

Cryptography, Blockchain and Coding Theory 2026
Lectures for high-school students

LEONARDO ERRATI

Cryptography Researcher

EXPERIENCE

Feb 2025/Now	Researcher in Cryptography and Number Theory , Politecnico di Torino, IT <i>PhD researcher; supervisors: Carlo Sanna, Edoardo Persichetti. Skills: methods in algebra and cryptography, scientific writing, organisation of self and groups.</i>	[link]
2024/2025	Research Fellow , Università di Trento, IT	[link]
2023	R&D Intern , Argentea SRL, IT <i>Restructured proprietary software architecture and coordinated cryptographic protocol updates. Skills: software architecture, protocol design, technical coordination.</i>	[link]

Scientific Appointments

Jul 2026/Now	Member of the Scientific Board , De Cifris APS	[link]
Jan 2026/Now	Member of the Scientific Board , Osservatorio Bitcoin	[link]

EDUCATION

Feb 2025/Now	PhD in Mathematics , Politecnico di Torino, IT <i>Cryptography and Number Theory</i>	[link]
Mar 2024	M.Sc. in Mathematics , Università di Trento, IT <i>Curriculum in Cryptography and Algebra</i>	[link]
Nov 2021	B.Sc. in Mathematics , Università di Trento, IT	[link]

SERVICE

Event Organisation

May 2026	CrypTography Days 2026 , Turin <i>Conference co-organiser</i>	[link]
Apr 2026	HisTO26 , History of Cryptography in Turin <i>Seminar co-organiser</i>	[link]
Sep 2025	Cifris25 , Various workshops <i>'A Journey into Bitcoin's Cryptography', 'Quantum RNGs & Cryptography', 'History of Cryptography'</i>	[link]

Editorial Service

May 2025	Bitcoin & Blockchain , De Cifris Koine, vol. 6 <i>With A. Bracciali and N. Gavioli</i>	[link]
Feb 2025	Proceedings of Cifris24 , De Cifris Koine, vol. 5 <i>With G. Cavicchioni, L. De Feo, B. Masucci, and M. Sala</i>	[link]

RESEARCH OUTPUT

Jul 2026	On Canonical Liftings, Group Actions, and Linkable Ring Signatures <i>With A. Budroni, G. D'Alconzo, and E. Persichetti</i>	[link]
Oct 2025	Design, Efficiency and Decentralised Applications of Threshold Schnorr Signatures <i>With M. Cenk and A. Meneghetti</i>	[link]

SELECTED TALKS

Sep 2026	ALPEC , Contributed talk <i>Canonical Liftings for Group Actions: Beyond Ring Signatures?</i>	[link]
Feb 2026	ITASEC26 , Talk <i>What the Pell: Cryptanalysis of Degree-two Pell Curves</i>	[link]
Sep 2025	Cifris25 , Talk <i>Cryptographic Standards for Random Number Generation</i>	[link]
Mar 2025	XRP Commons Paris , Invited talk <i>Post-Quantum Signatures for Blockchain</i>	[link]

SELECTED PROJECTS

Aug 2025	'Peano' Beamer Theme , PoliTo Cryptography Group <i>Semi-official Beamer theme for presentations.</i>	[link]
Apr 2021	'BeUnitn' Beamer Theme , UniTN Math Department <i>Semi-official Beamer theme for the Math Department.</i>	[link]